

Sales Performance Analysis of a Retail Chain



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Stock ID: 47857043



Introduction

Introduction

- Business Background: A globally operating retail chain
- Objective: Analyze sales performance, identify trends, and provide optimization recommendations
- Data Source: Kaggle retail sales dataset (cleaned and processed)

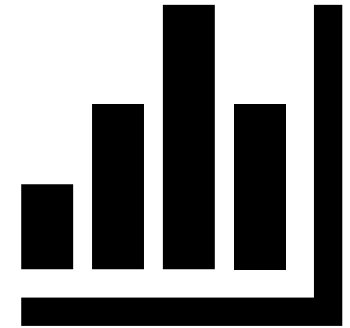




Methodology

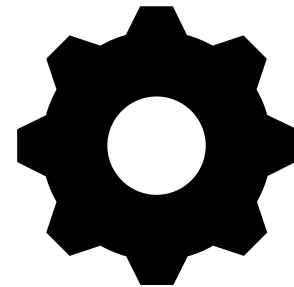
Data Details

- sales.csv: Daily sales data
- product_hierarchy.csv: Product hierarchy
- product_names.csv: Product names
- store_cities.csv: Store-to-city mapping
- store_names.csv: Store names
- city_names.csv: City names



Module 1: Data Cleaning and Preparation Using Excel

Used Excel for data cleaning: handled missing values (deleted rows / filled with None), removed duplicates, formatted dates and numbers, applied TRIM to remove extra spaces; used pivot tables to summarize sales data.



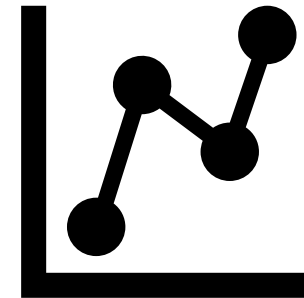
Module 2: Data Querying and Analysis Using PostgreSQL

Used PostgreSQL and pgAdmin to create the database, import data, write SQL queries for sales analysis, and perform multidimensional analysis using ROLLUP and CUBE.



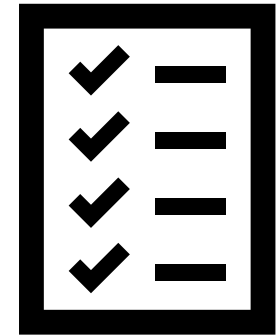
Module 3: Data Visualization and Statistical Analysis

Used Excel to create line charts, bar charts, sunburst charts, and scatter plots; used regression analysis to evaluate the impact of date on sales revenue.



Module 4: Data Visualization and Dashboards Using Tableau

Used Tableau to create four charts and assembled them into an interactive dashboard, added filter actions to enable cross-chart interactivity.





Results

Module 1, Lesson 1: Data Cleaning and Preparation

- Data cleaning ensured data quality and laid the foundation for subsequent analysis. Missing values were handled (promo_bin_1 filled with None), duplicates were removed, and date and number formats were standardized.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	product_id	store_id	date	sales	revenue	stock	price	promo_type_1	promo_bin_1	promo_type_2	promo_bin_2	promo_discount_2	promo_discount_type_2
2	P0001	S0002	2002/1/17	0	0	8	6.25	PR14	None	PR03	None		0 No discount
3	P0001	S0012	2002/1/17	1	5.3	0	6.25	PR14	None	PR03	None		0 No discount
4	P0001	S0013	2002/1/17	2	10.59	0	6.25	PR14	None	PR03	None		0 No discount
5	P0001	S0023	2002/1/17	0	0	6	6.25	PR14	None	PR03	None		0 No discount
6	P0001	S0025	2002/1/17	0	0	1	6.25	PR14	None	PR03	None		0 No discount
7	P0001	S0027	2002/1/17	0	0	7	6.25	PR14	None	PR03	None		0 No discount
8	P0001	S0040	2002/1/17	0	0	19	6.25	PR14	None	PR03	None		0 No discount
9	P0001	S0049	2002/1/17	0	0	8	6.25	PR14	None	PR03	None		0 No discount
10	P0001	S0050	2002/1/17	0	0	5	6.25	PR14	None	PR03	None		0 No discount
11	P0001	S0051	2002/1/17	0	0	6	6.25	PR14	None	PR03	None		0 No discount
12	P0001	S0055	2002/1/17	0	0	6	6.25	PR14	None	PR03	None		0 No discount
13	P0001	S0056	2002/1/17	1	5.3	6	6.25	PR14	None	PR03	None		0 No discount
14	P0001	S0062	2002/1/17	0	0	2	6.25	PR14	None	PR03	None		0 No discount
15	P0001	S0063	2002/1/17	0	0	7	6.25	PR14	None	PR03	None		0 No discount
16	P0001	S0066	2002/1/17	0	0	1	6.25	PR14	None	PR03	None		0 No discount
17	P0001	S0078	2002/1/17	0	0	6	6.25	PR14	None	PR03	None		0 No discount
18	P0001	S0082	2002/1/17	0	0	4	6.25	PR14	None	PR03	None		0 No discount
19	P0001	S0083	2002/1/17	0	0	10	6.25	PR14	None	PR03	None		0 No discount
20	P0001	S0087	2002/1/17	0	0	10	6.25	PR14	None	PR03	None		0 No discount
21	P0001	S0094	2002/1/17	0	0	6	6.25	PR14	None	PR03	None		0 No discount
22	P0001	S0096	2002/1/17	0	0	2	6.25	PR14	None	PR03	None		0 No discount
23	P0001	S0102	2002/1/17	0	0	7	6.25	PR14	None	PR03	None		0 No discount
24	P0001	S0103	2002/1/17	1	5.3	10	6.25	PR14	None	PR03	None		0 No discount
25	P0001	S0104	2002/1/17	0	0	22	6.25	PR14	None	PR03	None		0 No discount
26	P0001	S0105	2002/1/17	0	0	6	6.25	PR14	None	PR03	None		0 No discount
27	P0001	S0106	2002/1/17	1	5.3	3	6.25	PR14	None	PR03	None		0 No discount
28	P0001	S0107	2002/1/17	0	0	2	6.25	PR14	None	PR03	None		0 No discount
29	P0001	S0108	2002/1/17	0	0	7	6.25	PR14	None	PR03	None		0 No discount
30	P0001	S0110	2002/1/17	0	0	1	6.25	PR14	None	PR03	None		0 No discount
31	P0001	S0115	2002/1/17	0	0	1	6.25	PR14	None	PR03	None		0 No discount
32	P0001	S0122	2002/1/17	0	0	1	6.25	PR14	None	PR03	None		0 No discount
33	P0001	S0125	2002/1/17	0	0	7	6.25	PR14	None	PR03	None		0 No discount
34	P0001	S0131	2002/1/17	0	0	2	6.25	PR14	None	PR03	None		0 No discount

< >

sales

product_hierarchy

store_cities

store_names

city_names

product_names

Sheet1

+

Module 1, Lesson 2: Data Analysis Using Pivot Tables

Helps the business identify high/low performing stores, regional differences, and best/worst selling products.

	A	B	C	D	E
3	行标签	总销量	总库存	总收入	平均销量
4	AGD Centrum	129	4683	303.74	0.320099256
5	AGD Expert	29	5812	252.2	0.06401766
6	Appliance Centre	64	3400	122.37	0.297674419
7	Appliance Direct	89	6406	197.62	0.22997416
8	Appliance Paradise	114	2223	253.02	0.513513514
9	Appliance World	223	5929	755.83	0.487964989
10	Bimeks	158	4894	313.17	0.421333333
11	Block	946	16626	2864.18	1.261333333
12	Boulangier (National Chain)	105	6547	275.34	0.207100592
13	Casa da Tecnologia	36	4050	72.14	0.09
14	Casa Digital	200	17073	695.1	0.180342651
15	Casa Tech	323	10981	897.21	0.569664903
16	Cecchini	81	2484	152.3	0.487951807
17	Centro delloElettronica	144	4630	363.46	0.348668281
18	Comet	92	4860	276.27	0.227722772
19	Conrad Electronic (National Chain)	39	2673	73.59	0.176470588
20	Coolblue	402	5383	1289.11	0.899328859
21	Cosmos Sport	72	5471	197.89	0.143141153
22	Currys	849	35078	1673.98	0.907051282
23	Currys (National Chain)	83	7415	194.22	0.144347826
24	Darty	200	6727	414.75	0.429184549
25	Darty (National Chain)	90	2964	122.12	0.532544379
26	Datart (National Chain)	104	2200	188.36	0.495238095
27	DIGI	335	10397	875.68	0.480631277
28	DigiLife	249	5924	811.85	0.577726218
29	Dixons	259	10860	775.03	0.354309166
30	DNS (National Chain)	627	20939	2286.67	0.769325153
31	Dustin	130	9547	325.72	0.216306156
32	El Corte Ingls	165	7728	426.75	0.294117647
33	El Corte Ingls (National Chain)	233	7194	596.31	0.44980695
34	El Gigante	275	6700	891.05	0.609756098
35	Eldorado (National Chain)	926	14569	4264.16	1.770554493
36	Electro Dz	142	3172	254.88	0.689320388
<	sales	product_hierarchy	store_cities	store_names	city

Sales data analysis

	A	B	C	D
3	行标签	总销量	总收入	平均销量
4	Barcelona	109	209.53	0.251732102
5	Belgrade	599	1409.15	0.374141162
6	Berlin	505	1119.53	0.642493639
7	Brussels	337	652.34	0.382086168
8	Budapest	849	1673.98	0.907051282
9	Copenhagen	275	596.66	0.353470437
10	Dublin	408	843.64	0.249084249
11	Edinburgh	335	875.68	0.480631277
12	Florence	469	1381.19	0.283898305
13	Frankfurt	72	197.89	0.143141153
14	Geneva	83	197.59	0.168016194
15	Helsinki	1383	3110.87	0.38512949
16	Istanbul	326	796.99	0.493192133
17	Krakow	165	427.57	0.321011673
18	Kyiv	152	364	0.382871537
19	Lisbon	198	494.08	0.319354839
20	London	9174	31311.85	0.740854397
21	Luxembourg	204	683.62	0.167350287
22	Madrid	282	856.52	0.370565046
23	Milan	253	794.55	0.328145266
24	Munich	306	882.5	0.427374302
25	Oslo	167	403.07	0.269789984
26	Paris	135	260.09	0.535714286
27	Prague	330	1107.16	0.306122449
28	Reykjavik	217	608.32	0.302228412
29	Riga	781	3001.9	0.351485149
30	Rome	242	523.12	0.580335731
31	Saint Petersburg	2537	6760.03	0.417475728
32	Tallinn	39	88.13	0.176470588
33	Venice	769	2121.79	0.443227666
34	Vienna	744	1879.84	0.303302079
35	Warsaw	190	520.43	0.271816881
36	Zurich	261	863.77	0.341176471
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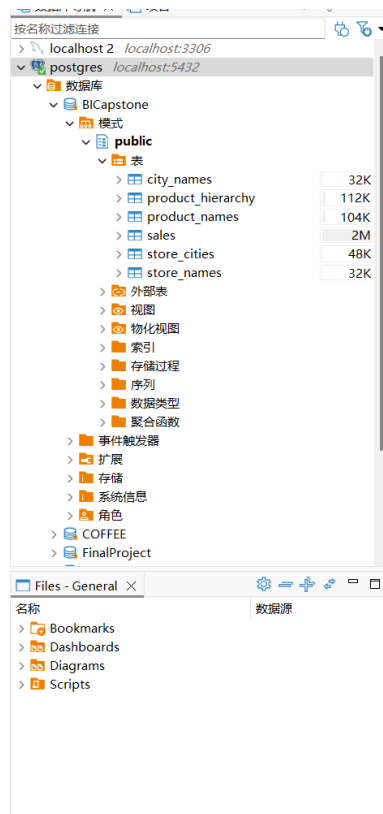
Sales by city analysis

	A	B	C	D
3	行标签	总销量	总收入	平均销量
4	Air Purifier	0	0	0
5	AutoBlendercast	0	0	0
6	AutoBlenderhub	272	751.96	0.852664577
7	AutoBlenderwave	5	105.7	0.02994012
8	AutoDryerdrive	18	179.11	0.059405941
9	AutoFridgeiron	1	23.69	0.012820513
10	AutoGrillflow	17	31.45	0.354166667
11	AutoHeatercast	81	373.13	0.35840708
12	AutoHeaterdrive	13	39.13	0.068062827
13	AutoHeatergen	2	25.26	0.071428571
14	AutoHeatermatic	4	36.86	0.333333333
15	AutoMixerlux	5	82.19	0.555555556
16	AutoTVdrive	53	397.5	0.148876404
17	AutoTVflow	0	0	0
18	AutoVacuumcast	1	70.76	0.055555556
19	AutoVacuumdrive	10	49.77	0.277777778
20	AutoVacuumhub	9	114.33	0.096774194
21	AutoVacuummatic	0	0	0
22	AutoWashercast	0	0	0
23	AutoWasherflow	14	183.95	0.066666667
24	AutoWasherlux	3	27.84	0.076923077
25	Baby Monitors with Video and Temperature Monitoring	3	24	0.017857143
26	Beverage Center	3	11.43	0.15
27	Bottom Freezer Refrigerator with Water Dispenser	29	158.39	0.163841808
28	Bread Machine with Gluten-Free Setting	68	243.13	0.618181818
29	Built-In Dishwasher with Third Rack	2	9.16	0.25
30	Canister Vacuum Cleaner	105	150.94	0.37366548
31	Canister Vacuums with HEPA Filter and Variable Suction Control	13	19.27	0.08125
32	Carpet Cleaners	0	0	0
33	Charcoal Grills with Smoker Box and Temperature Gauge	5	18.05	0.625
34	Chest Freezer with Lock and Key	2	40.5	0.018518519
35	Convection Oven	62	137.07	0.196202532
36	Counter-depth Refrigerator	194	404.01	0.45754717
<	Sheet3	Sheet6	sales	product_hierarchy
			store_cities	store_names
				city_names

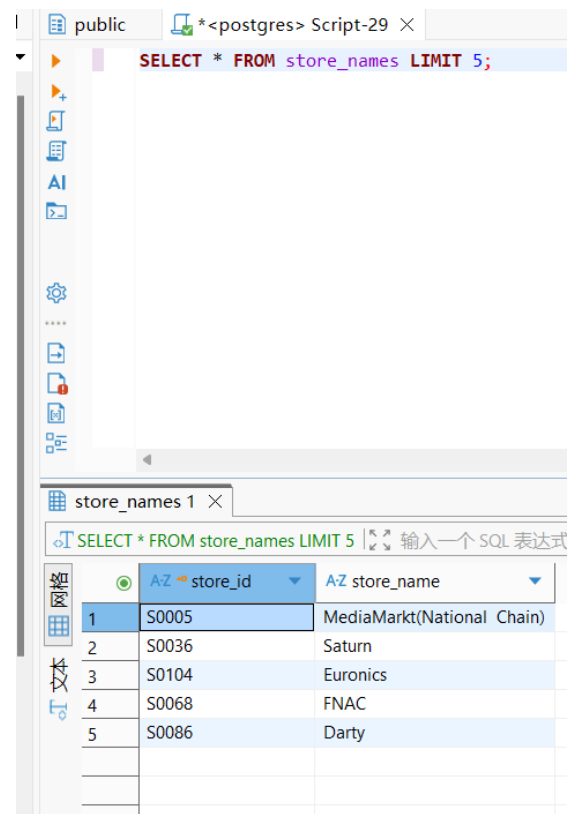
Product performance analysis

Module 2, Lesson 1: Data Querying Using PostgreSQL

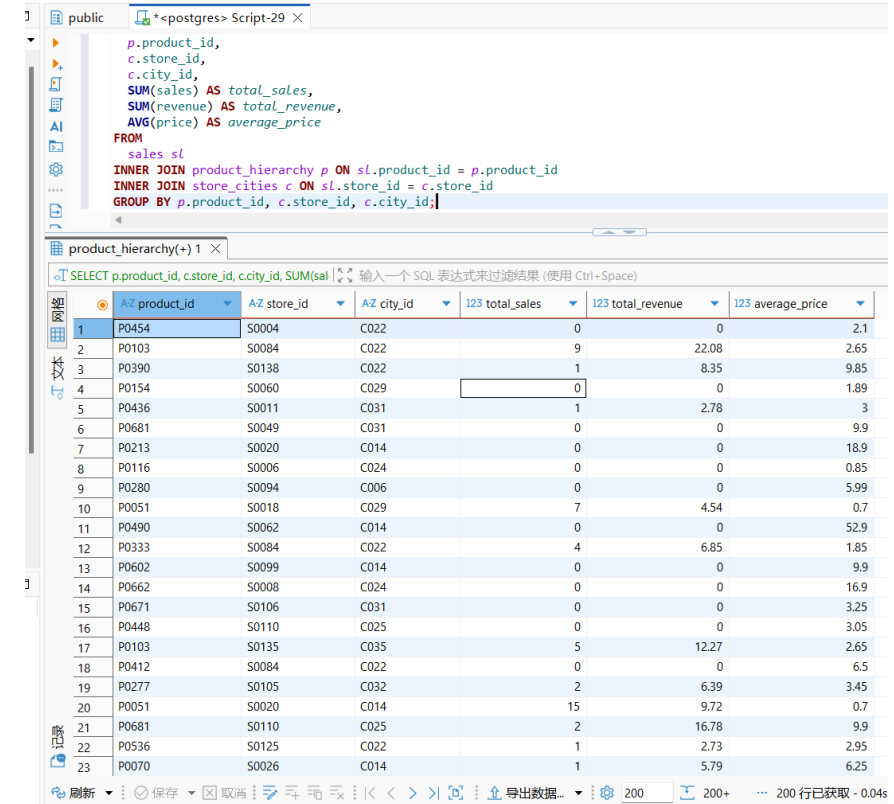
The database setup enables efficient querying and complex analysis of large-scale data.



Six tables under the BICapstone database



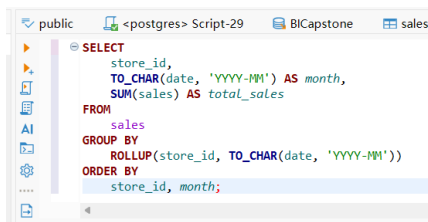
SQL query and results for data population verification



SQL query and results for sales performance analysis

Module 2, Lesson 2: Data Analysis Using PostgreSQL

Multidimensional analysis helps optimize sales strategies and inventory management.



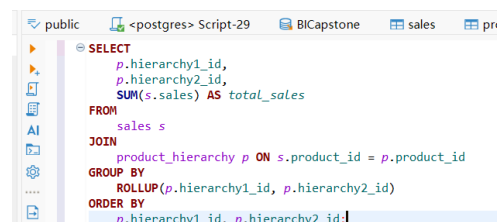
```
SELECT
    store_id,
    TO_CHAR(date, 'YYYY-MM') AS month,
    SUM(sales) AS total_sales
FROM
    sales
GROUP BY
    ROLLUP(store_id, TO_CHAR(date, 'YYYY-MM'))
ORDER BY
    store_id, month;
```

sales 1 X

SELECT store_id, TO_CHAR(date, 'YYYY-MM') AS month, SUM(sales) AS total_sales

	AZ store_id	AZ month	123 total_sales
1	S0001	2017-02	67.695
2	S0001	2017-03	35
3	S0001	[NULL]	102.695
4	S0002	2017-02	73.965
5	S0002	2017-03	28
6	S0002	[NULL]	101.965
7	S0003	2017-02	32
8	S0003	2017-03	20.835
9	S0003	[NULL]	52.835
10	S0004	2017-02	31
11	S0004	2017-03	8
12	S0004	[NULL]	39
13	S0006	2017-02	8
14	S0006	2017-03	4
15	S0006	[NULL]	12
16	S0008	2017-02	16
17	S0008	2017-03	15
18	S0008	[NULL]	31
19	S0009	2017-02	9
20	S0009	2017-03	13
21	S0009	[NULL]	22
22	S0010	2017-02	103.175
23	S0010	2017-03	37
24	S0010	[NULL]	140.175

Created data cubes using ROLLUP



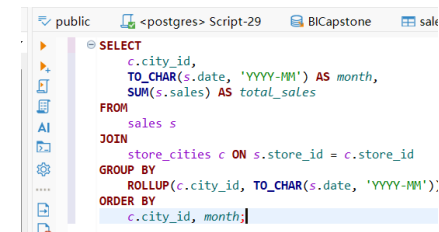
```
SELECT
    p.hierarchy1_id,
    p.hierarchy2_id,
    SUM(s.sales) AS total_sales
FROM
    sales s
JOIN
    product_hierarchy p ON s.product_id = p.product_id
GROUP BY
    ROLLUP(p.hierarchy1_id, p.hierarchy2_id)
ORDER BY
    p.hierarchy1_id, p.hierarchy2_id;
```

product_hierarchy 1 X

SELECT p.hierarchy1_id, p.hierarchy2_id, SUM(s.sales) AS total_sales

	AZ hierarchy1_id	AZ hierarchy2_id	123 total_sales
1	H00	H0000	1,380
2	H00	H0001	617
3	H00	H0002	849
4	H00	H0003	4,268
5	H00	H0004	613.366
6	H00	[NULL]	7,727.366
7	H01	H0105	175
8	H01	H0106	166
9	H01	H0107	374
10	H01	H0108	172
11	H01	[NULL]	887
12	H02	H0209	43.21
13	H02	H0210	182.305
14	H02	[NULL]	225.515
15	H03	H0311	17
16	H03	H0312	222
17	H03	H0313	581
18	H03	H0314	190
19	H03	H0315	5
20	H03	H0316	0
21	H03	H0317	2
22	H03	[NULL]	1,017

Summarized data along hierarchies



```
SELECT
    c.city_id,
    TO_CHAR(s.date, 'YYYY-MM') AS month,
    SUM(s.sales) AS total_sales
FROM
    sales s
JOIN
    store_cities c ON s.store_id = c.store_id
GROUP BY
    ROLLUP(c.city_id, TO_CHAR(s.date, 'YYYY-MM'))
ORDER BY
    c.city_id, month;
```

store_cities 1 X

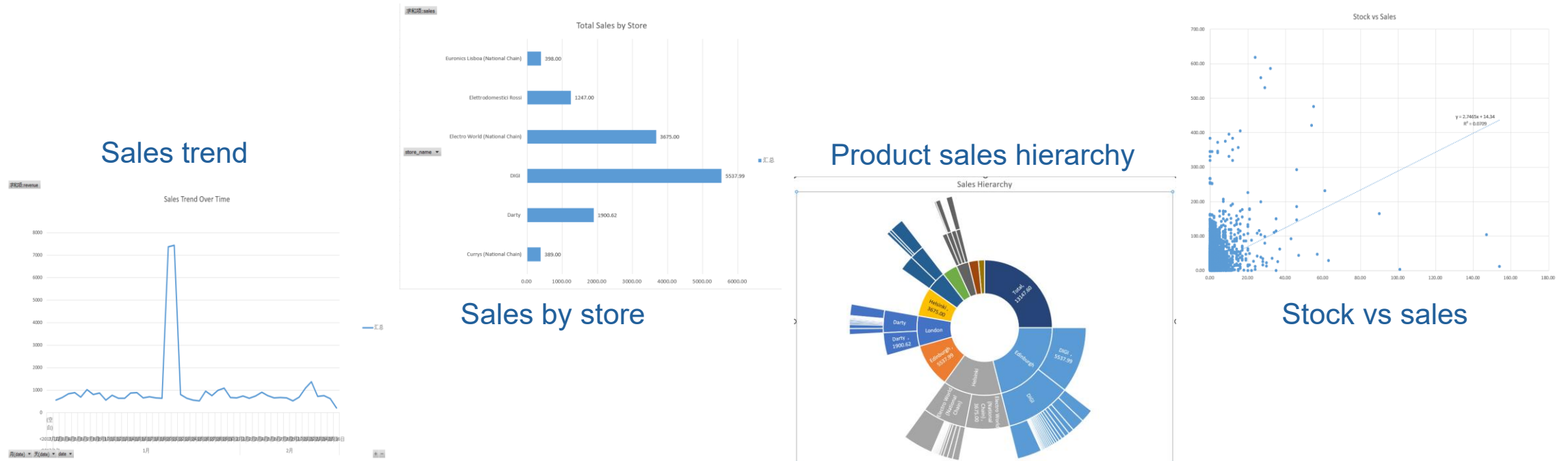
SELECT c.city_id, TO_CHAR(s.date, 'YYYY-MM') AS month, SUM(s.sales) AS total_sales

	AZ city_id	AZ month	123 total_sales
1	C002	2017-02	100
2	C002	2017-03	59
3	C002	[NULL]	159
4	C003	2017-02	32.84
5	C003	2017-03	14
6	C003	[NULL]	46.84
7	C004	2017-02	223.49
8	C004	2017-03	110.208
9	C004	[NULL]	333.698
10	C005	2017-02	54
11	C005	2017-03	35
12	C005	[NULL]	89
13	C006	2017-02	124
14	C006	2017-03	29
15	C006	[NULL]	153
16	C007	2017-02	73.965
17	C007	2017-03	28
18	C007	[NULL]	101.965
19	C008	2017-02	117.855
20	C008	2017-03	60
21	C008	[NULL]	177.855
22	C009	2017-02	21

Identified sales trends over time and across regions

Module 3, Lesson 1: Data Visualization Using Excel

Multiple chart types reveal data patterns from different perspectives.



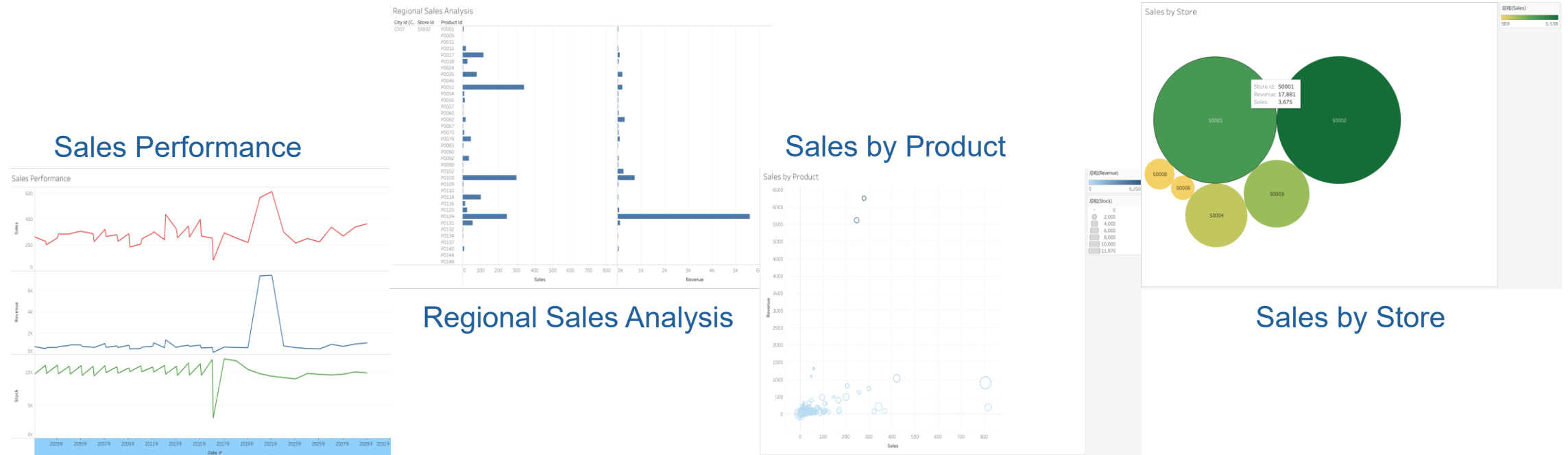
Module 3, Lesson 2: Statistical Analysis

- $R^2 \approx 0$, indicating that date does not explain the variation in sales revenue
- P-value = 0.9367 > 0.05, indicating that date does not have a significant impact on sales revenue
- Conclusion: Sales fluctuations are primarily influenced by other factors (promotions, pricing, inventory)

L	M	N	O	P	Q	R	S	T
promo_type	promo_type	promo_type	promo_type	promo_type	promo_type	promo_type	promo_type	promo_type
SUMMARY OUTPUT								
回归统计								
Multiple R	0.00056156							
R Square	3.1535E-07							
Adjusted R Sq	-4.969E-05							
标准误差	41.2764357							
观测值	20000							
方差分析								
	df	SS	MS	F	Significance F			
回归分析	1	10.7443842	10.7443842	0.00630634	0.93670536			
残差	19998	34071475.3	1703.74414					
总计	19999	34071486.1						
	Coefficients	标准误差	t Stat	P-value	Lower 95%	Upper 95%	下限 95.0%	上限 95.0%
Intercept	2.04331791	0.58594066	3.48724379	0.00048907	0.89482582	3.19181	0.89482582	3.19181
X Variable 1	-0.0017988	0.02265179	-0.0794125	0.93670535	-0.0461982	0.04260055	-0.0461982	0.04260055

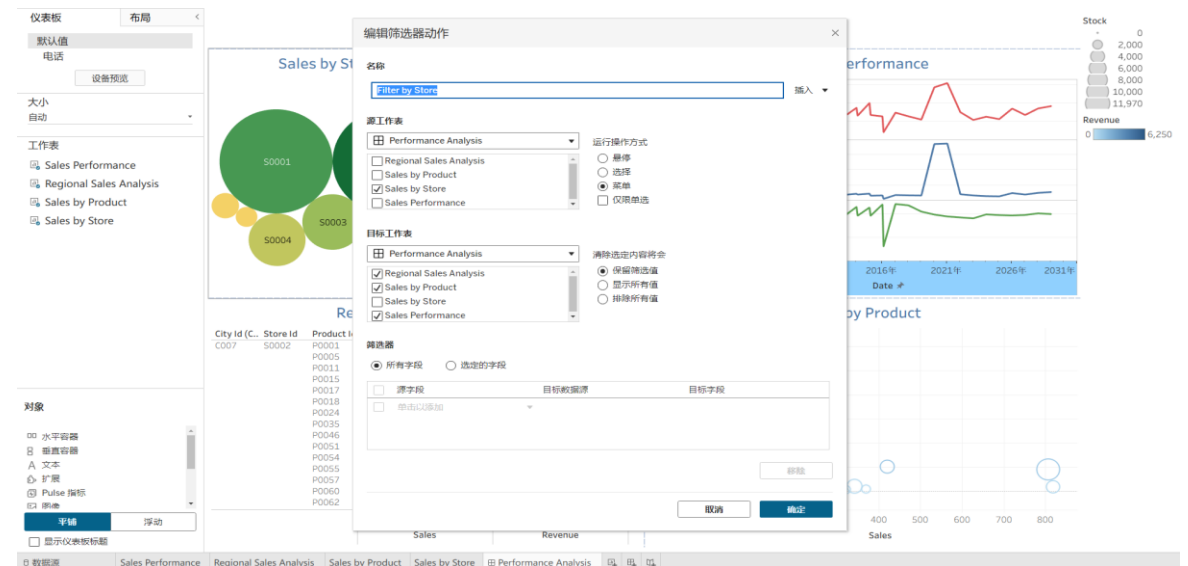
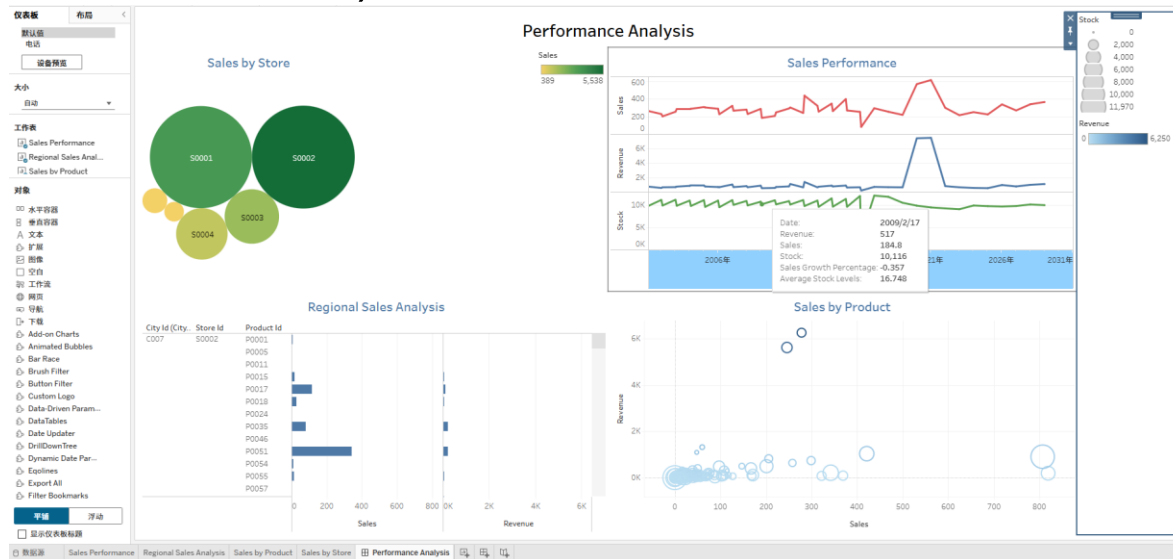
Module 4, Lesson 1: Basic Tableau Visualizations

Tableau visualizations clearly communicate sales insights and are easy for non-technical audiences to understand.



Module 4, Lesson 2: Advanced Visualizations Using Tableau

The interactive dashboard allows decision-makers to filter on demand (e.g., click on a store to view details).





Discussion

Insights and Recommendations

- Insight 1: Regression analysis shows that date does not significantly affect sales revenue, indicating no seasonal fluctuations in sales.
- Recommendation 1: Focus more on promotional activities and pricing strategies rather than seasonal adjustments.
- Insight 2: Some stores have sales significantly below the average.
- Recommendation 2: Analyze inventory, pricing, and promotions of low-performing stores and implement targeted optimizations.
- Insight 3: The sunburst chart shows that certain products have outstanding sales in specific cities.
- Recommendation 3: Increase distribution of best-selling products in advantageous regions and optimize inventory allocation.



Conclusion

Summary

- Completed a full analysis from data cleaning to visual dashboards.
- Found that sales are primarily influenced by promotions, pricing, and inventory, rather than time.
- The interactive dashboard can be used for daily sales monitoring and decision support.



Appendix

Appendix

